

ECON100B - SECTION 16

Intro to the IS-LM Model

Felix Zhang

ANNOUNCEMENTS/AGENDA

Announcements

- Regrade requests to be completed by end of next week

Agenda

- Recent Macro-clysms Part II (In uploaded deck)
- Introduction to the IS Curve

Questions of the Day

Post-Midterm Content Review:

1. Using the quantity theory of money and growth rates, if an economy has a 5% growth rate in money supply and a real GDP growth rate of 2%, what would the inflation rate equal?
2. If our short run output is \$21,000 and our Long run expected output is \$20,000, what is our short run output?
3. Suppose our short-run output is -2%. What does Okun's law state will happen to our cyclical unemployment rate?
4. Why did the collapse of mortgage-backed securities lead to the financial crisis in 2008? What caused the banking system to collapse?

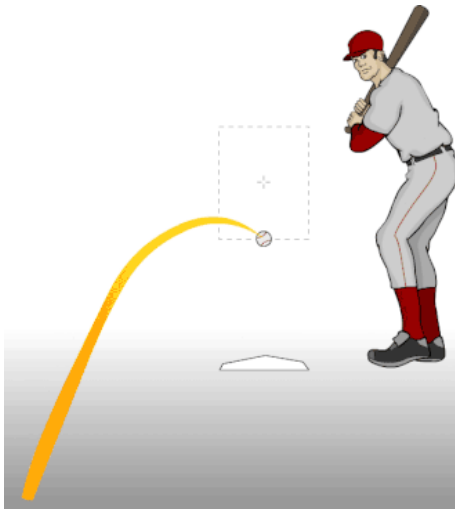
What word that isn't a (common) name would be a good name?

Questions of the Day Answers

Post-Midterm Content Review:

1. 3%
2. 5%
3. It would be -1%
4. See slides for further details. In very simple terms, the collapse occurred because banks held MBS, and due to the collapse of MBS, certain banks were unable to pay off debts to other banks, leading to a liquidity crisis.

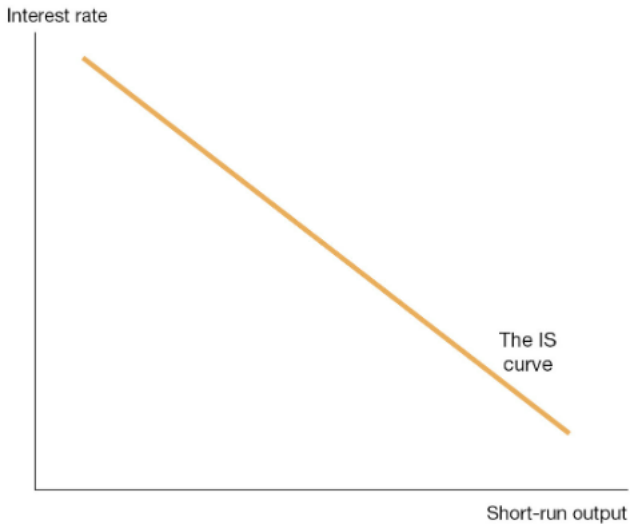
The IS Curve



Introduction to the IS Curve

- We previously discussed that the Federal Reserve affect the short-run economy by changing the interest rate – we'll formalize this through our IS curve
- The relationship that we observe is that increasing our interest rate decreases investment, which decreases our total output
 - Increasing the interest rate means firms reduce their spending of capital, while consumers reduce borrowing (for things like houses, loans, etc.)

The IS Curve



Setup of the IS curve

- We reintroduce our national income identity equation

$$Y_t = C_t + I_t + G_t + EX_t - IM_t.$$

$$C_t = \bar{a}_c \bar{Y}_t,$$

$$G_t = \bar{a}_g \bar{Y}_t,$$

$$EX_t = \bar{a}_{ex} \bar{Y}_t,$$

$$IM_t = \bar{a}_{im} \bar{Y}_t,$$

$$\frac{I_t}{\bar{Y}_t} = \bar{a}_i - \bar{b} \left(R_t - \bar{r} \right).$$

Assumptions of our model

- According to our model, even as actual GDP is affected, our potential output remains unchanged (exogenously given)
- This tells us that factors like consumption will not change either
- During times of recession, people might keep their consumption steady by drawing from savings
- The investment equation also depends on the gap between the real interest rate and the marginal product of capital
- If the marginal product of capital is low relative to the real interest rate, firms are better off saving their money rather than borrowing. If the marginal product of capital is high relative to the real interest rate, then firms are better off borrowing
- The parameter b then tells us how sensitive investment is to changes in the interest rate

IS Curve Questions

- If there is an increase in the real interest rate, how would this impact investment? Why?
- If we observed an increase in the parameter $\overline{a_g}$, what does this denote about government spending in relation to potential GDP?
- Suppose our real interest rate is 10% and our marginal product of capital is 12%. If a firm borrows 100 units at the 10% rate and invests it in capital, what would its predicted output be? What would its profit be?

IS Curve Answers

- This would decrease investment as the cost to borrow has gone up, thus making it a less attractive option
- The percentage of government spending in relation to potential GDP has gone up
- It's predicted output would be 12 units. It pays 10 units back on borrowing, and then its profit would be two units

Deriving the IS Curve

First we divide our actual output by our expected output

$$\frac{Y_t}{\bar{Y}_t} = \frac{C_t}{\bar{Y}_t} + \frac{I_t}{\bar{Y}_t} + \frac{G_t}{\bar{Y}_t} + \frac{EX_t}{\bar{Y}_t} - \frac{IM_t}{\bar{Y}_t}.$$

Which gives us the following form:

$$\frac{Y_t}{\bar{Y}_t} = \bar{a}_c + \bar{a}_i - \bar{b} \left(R_t - \bar{r} \right) + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im}$$

Recalling our short run output, we can solve for the short run output with our new equation

$$\tilde{Y}_t \equiv \frac{Y_t - \bar{Y}_t}{\bar{Y}_t}.$$

Deriving the IS Curve

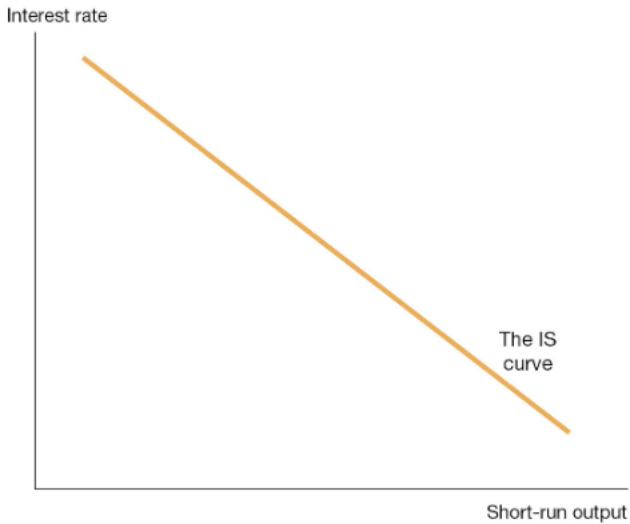
This gives us the following final equation

$$\tilde{Y}_t = \bar{a} - \bar{b}(R_t - \bar{r}),$$

Where:

$$\bar{a} \equiv \bar{a}_c + \bar{a}_i + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1,$$

The IS curve



The IS Curve Practice

- Using the IS curve, if there a decrease in $\bar{a}$, what will happen to short-run output?
- Using the IS curve, if the real interest rate equals the marginal product of capital and there are no shocks to aggregate demand, what will happen to short-run output?
- According to the IS curve, what will happen to short-run output if we have a high interest rate?

The IS Curve Practice Answers

- The short run output decreases as well, a shock to aggregate demand
- Short run output equals zero
- Short run output will decrease if the interest rate is higher